

Prashanth Sai Santosh Koripalli

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Education

Master of Science in Artificial Intelligence

Jan 2024 – May 2025

San Jose State University, USA

Course Work: AI and Data Engineering, Machine Learning, Intelligent Autonomous Systems, Deep Learning, Reinforcement Learning, Graphs in Robotics, Advance Data Mining, Enterprise Application Development

Technical Skills

Languages: Python, C++, C, Java, R, Rust, CUDA, Bash, JavaScript, ShellScript, TypeScript, SCSS, HTML/CSS, SQL.

Frameworks: PyTorch, TensorFlow, Ollama, Docker, Kubernetes, React, Next.js, Node, Redux, GraphQL.

Tools/ Paradigms: WandB, Git, Github, Jira, Dspy, OpenAI Gym, Jenkins, Power BI, Tableau, Amazon Web Services.

Databases: MySQL, OracleDB, SQL, PostgreSQL, MongoDB, Firebase, Apache Kafka, Redis.

Experience

Software Developer | Oracle Corporation

Aug 2022 – Jan 2024

- Built scalable ML inference pipelines to transition models from AWS to OCI, and enhanced OCI dashboards to support real-time risk ranking and visualization, improving decision-making for customers like the Red Bull Racing and Pfizer.
- Integrated Newmetrix's AI-powered safety risk models into Oracle Construction Intelligence Cloud (CIC) post-acquisition, enabling predictive analytics and automated safety insights for construction clients.
- Optimized PL/SQL scripts and SQL queries for Primavera Analytics, resolving Data Warehousing issues with sources like P6, Unifier, and tools like OBI and OCI, improving query speed and system reliability by 20%.

Data Analyst Intern | Oracle Corporation

Jan 2022 – July 2022

- Acquired expertise in Oracle tools like Primavera, P6, and OCI within 3 months by actively engaging in training, concurrently working on tasks, and manually testing daily runs, resulting in enhanced tool reliability and project efficiency.
- Automated daily and weekly test runs using Selenium for Oracle's analytics platforms, streamlining UI/UX workflows and functional testing, reducing manual testing time by 50% and improving testing efficiency.
- Led the successful migration of 15,000 Confluence documentations from CEGBU to GBU, ensuring no data loss by developing an automated script to validate and cross-check content, resulting in streamlined access and better collaboration across teams.

Projects

SAMMY AI Chatbot | Python, NLP, RAG, LLM, LangChain, LlamaIndex, LangFuse

- Built 'Sammy,' an LLM-powered chatbot for SJSU using a Retrieval-Augmented Generation (RAG) pipeline implemented with LangChain and LlamaIndex, enabling 24/7 query resolution with 100% positive feedback.
- Integrated LangFuse for real-time LLM monitoring, logging, and performance analysis to maintain response quality.
- Engineered a confidence-based response modulation system, dynamically adjusting the chatbot's tone and reducing 99% of model hallucinations in query responses.

RoboGrasp: Learning Manipulation via Human Demonstrations | Python, PyTorch, ROS2, OpenAI Gym, YOLOv8

- Engineered an imitation learning pipeline enabling a 6-DoF robotic arm to acquire generalizable manipulation policies from a limited set of kinesthetic demonstrations across diverse household object categories.
- Integrated YOLOv8 for instance-level object localization and trained a behavior cloning policy using vision-action trajectories in ROS2, achieving a 78% success rate on unseen object configurations in simulation-to-real transfer.

EBay Challenge Entity Extraction (Ebay ML Competition) | Python, Ollama, Dspy, PyTorch, TorchTune

- Fine-tuned Llama 3.1 with LoRA to map 5M+ automobile part records to compatible car models in Year-Make-Model format, utilizing 300GB of website data for improved accuracy.
- Designed a distributed inference system across multiple GPU nodes on HPC, cutting inference time from 10 days (single A100 GPU) to 23 hours using 16 A100 GPUs.
- Applied entropy-based sampling to filter high-confidence predictions, increasing the F0.2 score by 13% (0.49 to 0.62).

Perception & Path-Planning System | PyTorch, TensorRT, WandB, Distributed Training, Autonomous Systems

- Developed an object detection pipeline for RGB Camera with TensorRT, achieving 98% accuracy at 35 FPS.
- Built a reinforcement learning-based path-planning module, maintaining 90% speed and lane compliance in urban areas.
- Optimized multi-GPU training with MLOps, cutting training time by 40% while using WandB for tracking.

Achievements

- **Achieved 3rd Rank in eBay Machine Learning Challenge (2024), USA:** Successfully tackled entity extraction for mapping 5M+ automobile part records to compatible car models.
- **Massachusetts Institute of Technology Covid Challenge, Winning Team (2020):** Won the hackathon by developing a digital loan approval system for unregistered MSMEs to address key challenges faced by small businesses.